



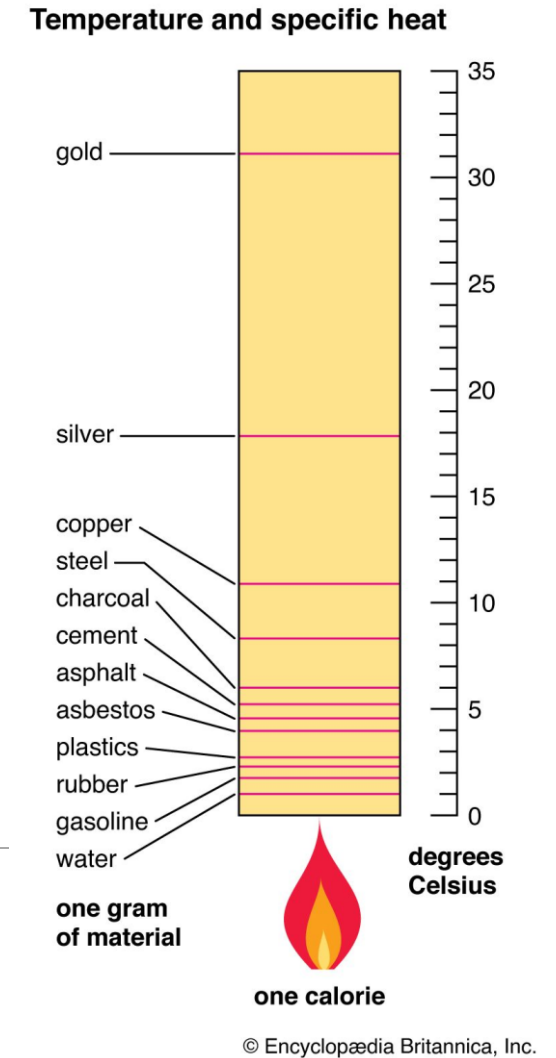
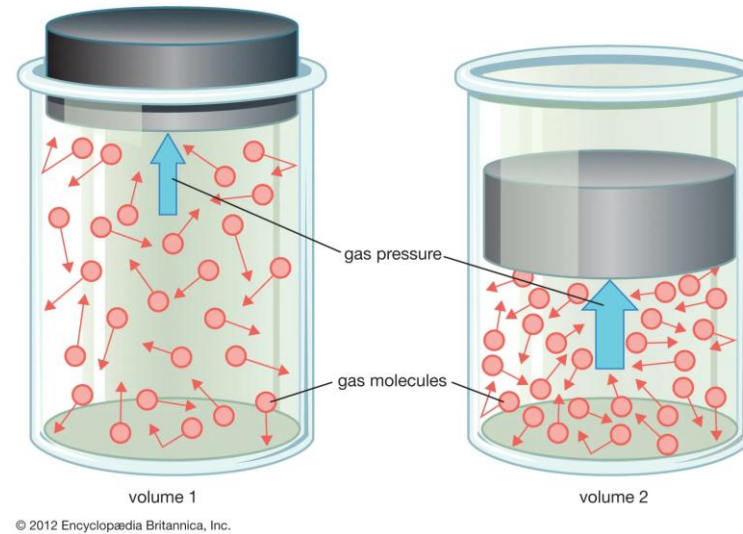
ME 2134

Thermodynamics

INSTRUCTOR: DR. JUSTIN DUBIK

SPRING 2026 – WEEK 3

IDEAL GAS LAW, COMPRESSIBLE GASES, ENTHALPY, SPECIFIC HEAT



(Re)introducing the ideal gas law

Vapor vs. gas

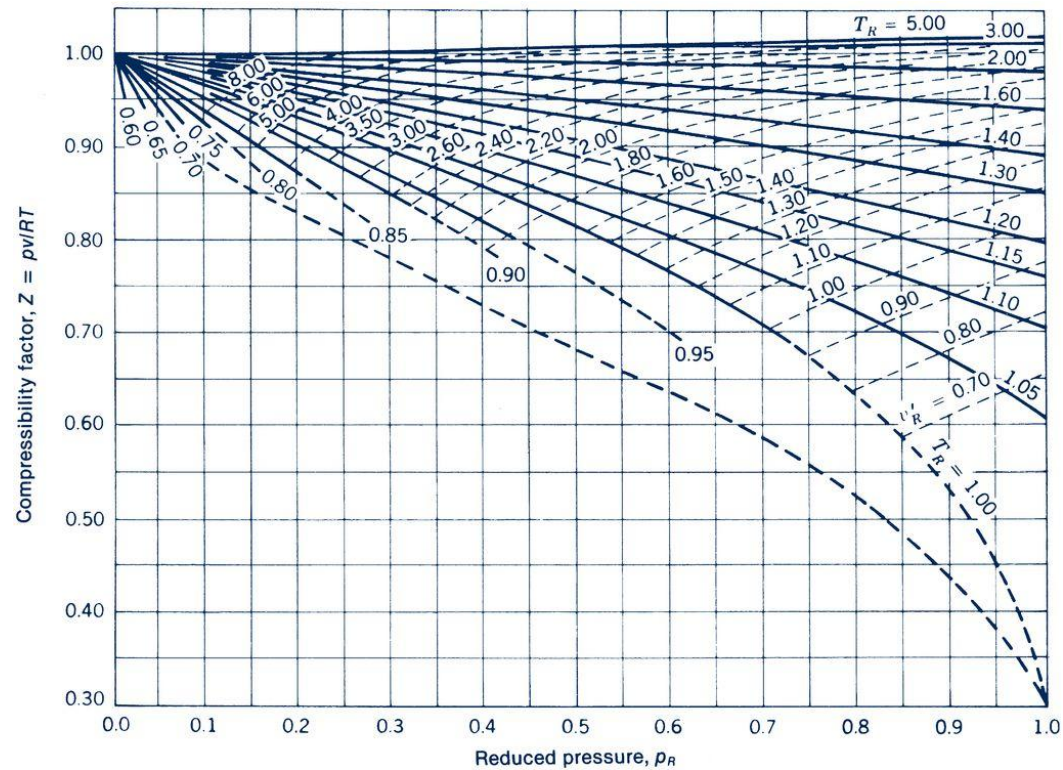
Let's try using the ideal gas law on vapor

Table 15.4.1: Properties of superheated water vapor (Table A-4).

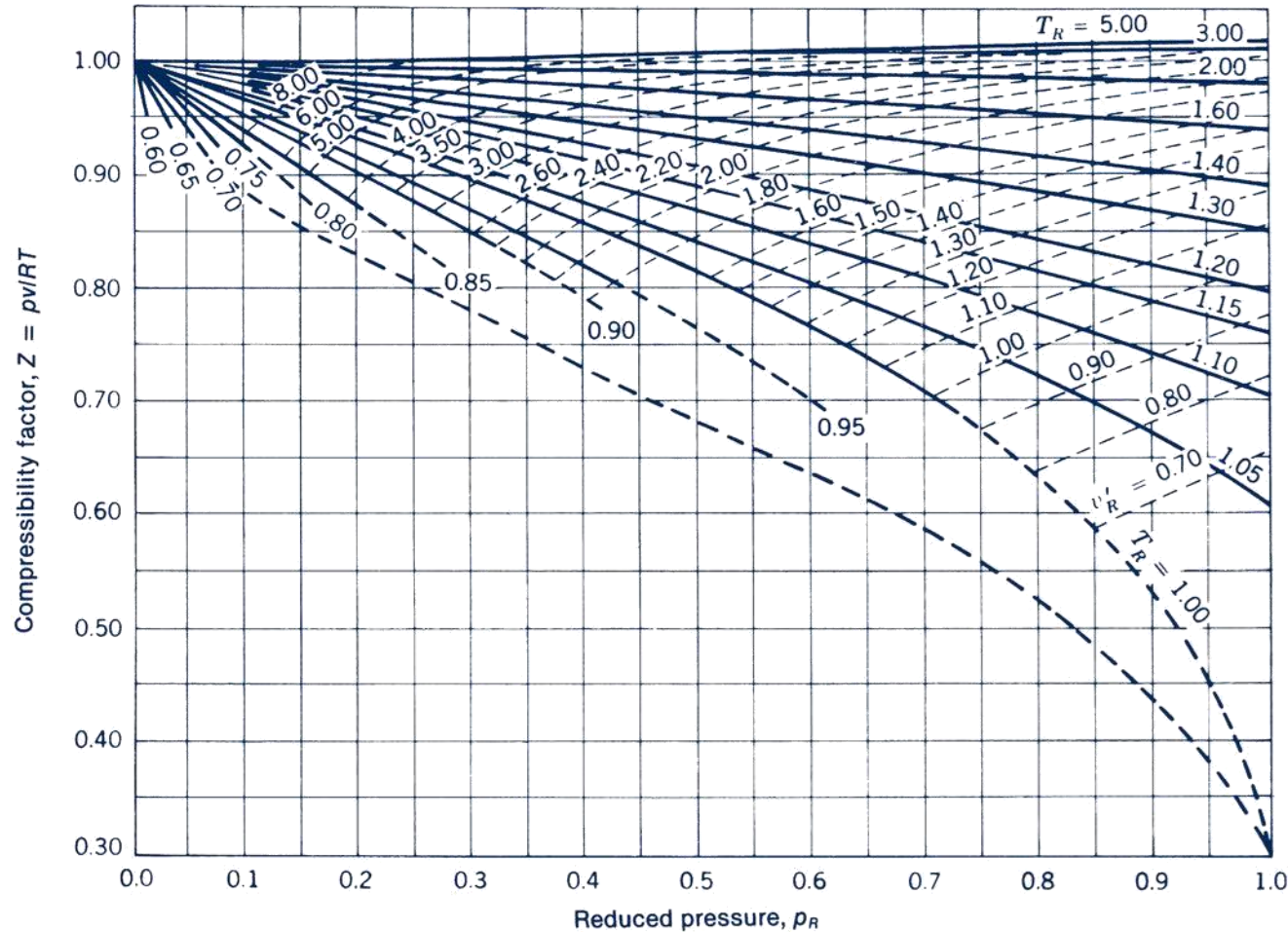
T °C	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg·K		v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg·K
$p = 20.0 \text{ bar} = 2.0 \text{ MPa}$ ($T_{\text{sat}} = 212.42^\circ\text{C}$)						$p = 30.0 \text{ bar} = 3.0 \text{ MPa}$ ($T_{\text{sat}} = 233.90^\circ\text{C}$)			
Sat.	0.0996	2600.3	2799.5	6.3409		0.0667	2604.1	2804.2	6.1869
240	0.1085	2659.6	2876.5	6.4952		0.0682	2619.7	2824.3	6.2265
280	0.1200	2736.4	2976.4	6.6828		0.0771	2709.9	2941.3	6.4462
320	0.1308	2807.9	3069.5	6.8452		0.0850	2788.4	3043.4	6.6245

The “ideal” gas

Modifying $Pv = RT$: the “Compressed gas law”



Compressibility example



- 5 kmol of oxygen (O_2) gas undergoes a process in a closed system from $P_1 = 50$ bar, $T_1 = 170$ K to $P_2 = 25$ bar, $T_2 = 200$ K.
- Determine the change in volume of the system

When to use compressed vs. ideal gas law

Seen h in tables... what is it?

Define H = enthalpy

Need to determine how to calculate/find h for various substances

Finding h

Table A-2, A-3

Enthalpy kJ/kg		
Sat. Liquid h_f	Evap. h_{fg}	Sat. Vapor h_g
0.01	2501.3	2501.4
16.78	2491.9	2508.7
20.98	2489.6	2510.6

Finding h for compressed liquids

Table A-5

T °C	$v \times 10^3$ m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg·K
$p = 25.0 \text{ bar} = 2.5 \text{ MPa}$ ($T_{\text{sat}} = 223.99^\circ\text{C}$)				
20	1.0006	83.80	86.30	.2961
40	1.0067	167.25	169.77	.5715
80	1.0280	334.29	336.86	1.0737
100	1.0423	418.24	420.85	1.3050
140	1.0784	587.82	590.52	1.7369
180	1.1261	761.16	763.97	2.1375
200	1.1555	849.9	852.8	2.3294
220	1.1898	940.7	943.7	2.5174
Sat.	1.1973	959.1	962.1	2.5546

Finding h for compressed liquids (cont.)

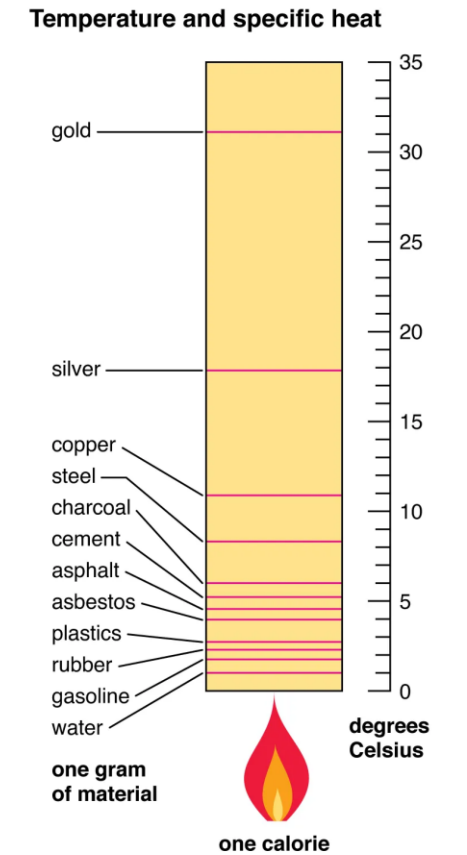
Table A-5

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Enthalpy example

- Supercooled water is at $P = 2500$ kPa and $T = 140$ °C
- Evaluate the specific enthalpy in two ways:
 - Using Table A-5
 - Using $h_{f@T}$
 - Using $h = h_{f@T} + v_{f@T}(P - P_{sat@T})$

Specific heat capacity



Two types of specific heat

Two types of specific heat (cont.)

An important note on our specific heats

Using specific heat of ideal gases

Specific heat example

- 1 kg of N_2 gas fills a piston cylinder.
- The piston cylinder expands from an initial state of $V_1 = 1 \text{ m}^3$ to $V_2 = 2 \text{ m}^3$, in an isobaric process (at a constant pressure) of $P = 1 \text{ atm}$
- Determine the heat transfer using a constant specific heat evaluated at 500 K

Specific heat of ideal gases (cont.)

Specific heat ratio

Polytropic processes (for all substances)

Polytropic processes for ideal gases

Specific heat & polytropic processes example

- 1 kg of O_2 gas expands from an initial state of $T_1 = 700$ K, $P_1 = 5$ bar to a final state where $P_2 = 2$ bar
- During the process, $PV^{1.3} = \text{constant}$
- Assume ideal gas behavior, negligible KE and PE
- Determine the heat transfer using a constant specific heat evaluated at 600 K

Specific heat of incompressible substances
